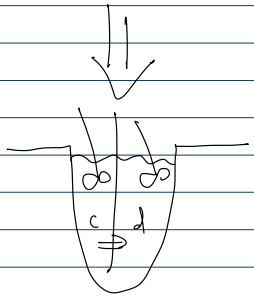


Reaction occuring in cont. phase (includes MT & reaction)

$\frac{d}{dt}[V_c c] = q_c c_f - q_c c - r_{rxn\ term}$ } for cont. phase

$\frac{d}{dt}[V_d c'] = q_d c'_f - q_d c' - MT$ to cont. phase } for dispersed phase

This sort of phase analysis is good only if you have very intense / fast mixing



Population balances are supposed to be a better approximate of this phase mixing.

Pop. balance:

$\int_0^{c_0} n\, dc_1$

$\bar{h} = \omega N_n$

$h^f = \omega\, dc_1 \iint_{c_1+c_2=2c} n(c_1,t)\, n(c_2,t)\, dc_2$

↳ Integrating over an area.
(But only integrating once (Not over a surface)

$\frac{\#}{time \cdot}$ Number density

$N|(c)$ → reciprocal of all internal variable

density in conc-space

Closed book:

Reaction invariance → where is it coming from (from Stoich)

Fresh alternative: $A_{\sim} x_{\sim} = b_{\sim}$

$A_{\sim} x_{\sim} = 0$ → If this has non-zero solution

$$\underset{\sim}{A}^T \underset{\sim}{\omega} = 0$$